

S28 - Probability: Total Probability & Bayes' Theorem

S28 ka focus: Bayes ko aur strong karna, especially **reverse probability**, **multiple groups**, aur **business interpretation**.

Tum answers bhejoge, main **sirf galti wali jagah par concept explain** karunga, kyunki ye topic abhi naya hai.

1 | Total Probability — Customer Conversion

An e-commerce company gets visitors from two sources:

Traffic Source	Visitors	Purchased
Advertisement	600	90
Organic Search	400	100

Answer:

1. Find $P(\text{Advertisement})$.
 2. Find $P(\text{Organic Search})$.
 3. Find $P(\text{Purchased} | \text{Advertisement})$.
 4. Find $P(\text{Purchased} | \text{Organic Search})$.
 5. Find the overall $P(\text{Purchased})$.
 6. Explain how the overall purchase probability is obtained from the two traffic sources.
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2 | Bayes — Job Applications

A company receives applications from three sources:

Source	Applications	Selected
LinkedIn	500	75
Job Portal	300	30
Referral	200	50

Answer:

1. Total applications.
 2. Total selected candidates.
 3. Find $P(\text{Selected})$.
 4. Find $P(\text{Selected} \mid \text{Referral})$.
 5. Find $P(\text{Referral} \mid \text{Selected})$.
 6. If a randomly selected candidate was selected, which source are they most likely to have come from?
 7. Is $P(\text{Selected} \mid \text{Referral})$ the same as $P(\text{Referral} \mid \text{Selected})$?
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3 | Bayes — Fraud Detection

A bank has the following transaction data:

Transaction Type	Transactions	Fraud
Online	8,000	160
Offline	2,000	40

Answer:

1. Total transactions.
2. Total fraudulent transactions.
3. $P(\text{Fraud} \mid \text{Online})$.

4. $P(\text{Fraud} \mid \text{Offline})$.
 5. $P(\text{Online} \mid \text{Fraud})$.
 6. If a transaction is known to be fraudulent, what is the probability that it was online?
 7. Which is higher: online fraud rate or offline fraud rate?
 8. Can we say that every online transaction is fraudulent? Explain.
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4 | Reverse Probability — Customer Churn

A subscription company has:

Plan	Customers	Churned
Monthly	1,200	240
Annual	2,800	140

Answer:

1. Total customers.
 2. Total churned customers.
 3. Overall churn probability.
 4. $P(\text{Churned} \mid \text{Monthly})$.
 5. $P(\text{Churned} \mid \text{Annual})$.
 6. $P(\text{Monthly} \mid \text{Churned})$.
 7. Which plan has the higher churn rate?
 8. Among churned customers, which plan contributes more churned customers?
 9. Are Q6 and Q4 asking the same question? Explain.
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5 | Total Probability — Marketing Campaign

A company runs two campaigns:

Campaign	Customers Reached	Purchased
Campaign A	3,000	240
Campaign B	2,000	300

Answer:

1. Total customers reached.
 2. Total purchasers.
 3. $P(\text{Purchased} \mid \text{Campaign A})$.
 4. $P(\text{Purchased} \mid \text{Campaign B})$.
 5. Overall purchase probability.
 6. Which campaign has the higher conversion rate?
 7. Which campaign generated more purchasers?
 8. Explain why the campaign with more purchasers does not necessarily have the higher conversion rate.
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6 | Bayes — Customer Support

A company receives support requests:

Channel	Requests	Resolved
Chat	700	560
Email	500	300

Answer:

1. Total requests.
2. Total resolved requests.
3. $P(\text{Resolved} \mid \text{Chat})$.

4. $P(\text{Resolved} \mid \text{Email})$.
 5. $P(\text{Chat} \mid \text{Resolved})$.
 6. If a request is known to be resolved, what is the probability it came from Chat?
 7. Which channel has the better resolution rate?
 8. Which channel contributes more resolved requests?
 9. Should the company automatically stop using Email? Why or why not?
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7 | Three-Group Bayes Case — Employee Skills

A company has:

Department	Employees	Python Skilled
Data Analytics	100	70
Finance	150	30
Marketing	250	25

Answer:

1. Total employees.
 2. Total Python-skilled employees.
 3. $P(\text{Python Skilled})$.
 4. $P(\text{Python Skilled} \mid \text{Data Analytics})$.
 5. $P(\text{Data Analytics} \mid \text{Python Skilled})$.
 6. If an employee is Python-skilled, which department are they most likely to belong to?
 7. Which department has the highest Python skill rate?
 8. Are Q6 and Q7 answering the same business question? Explain.
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8 | Find the Wrong Analyst Interpretation

An analyst says:

“The probability that a customer purchases given that they saw an advertisement is 25%. Therefore, if exactly 1,000 customers see the advertisement, exactly 250 customers will purchase.”

Answer:

1. Is the conclusion correct?
 2. What is wrong with the word “**exactly**”?
 3. What does 25% actually represent?
 4. Approximately how many customers might purchase?
 5. Is that number guaranteed?
 6. Give a better version of the analyst's statement.
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9 | Conditional vs Reverse Probability

A dataset contains 2,000 customers:

- 800 customers used the mobile app.
- 320 mobile-app users purchased.
- 500 customers purchased in total.

Answer:

1. Find $P(\text{App User})$.
2. Find $P(\text{Purchased})$.
3. Find $P(\text{Purchased} \mid \text{App User})$.
4. Find $P(\text{App User} \mid \text{Purchased})$.
5. Which probability answers: “**Among app users, how many purchased?**”
6. Which probability answers: “**Among purchasers, how many used the app?**”
7. Are Q3 and Q4 the same? Why?

10 | Full Data Analyst Case — Subscription Business

A subscription company analyzes 5,000 customers:

Customer Segment	Customers	Churned
Basic	2,500	375
Premium	1,500	90
Enterprise	1,000	35

Answer:

1. Calculate total churned customers.
2. Calculate overall churn probability.
3. Calculate $P(\text{Churned} \mid \text{Basic})$.
4. Calculate $P(\text{Churned} \mid \text{Premium})$.
5. Calculate $P(\text{Churned} \mid \text{Enterprise})$.
6. Calculate $P(\text{Basic} \mid \text{Churned})$.
7. Which segment has the highest churn rate?
8. Which segment contributes the largest number of churned customers?
9. Explain why Q7 and Q8 can have different answers.
10. If the company wants to **reduce the highest churn rate**, which segment should it investigate?
11. If the company wants to **reduce the largest number of churned customers**, which segment should it investigate?
12. Should management immediately conclude that the Basic plan is causing customers to churn? Explain.
13. Give one additional piece of data you would want before making a final business decision.